



LIANGSHAN ENTERPRISE

SECURITY ARCHITECTURE

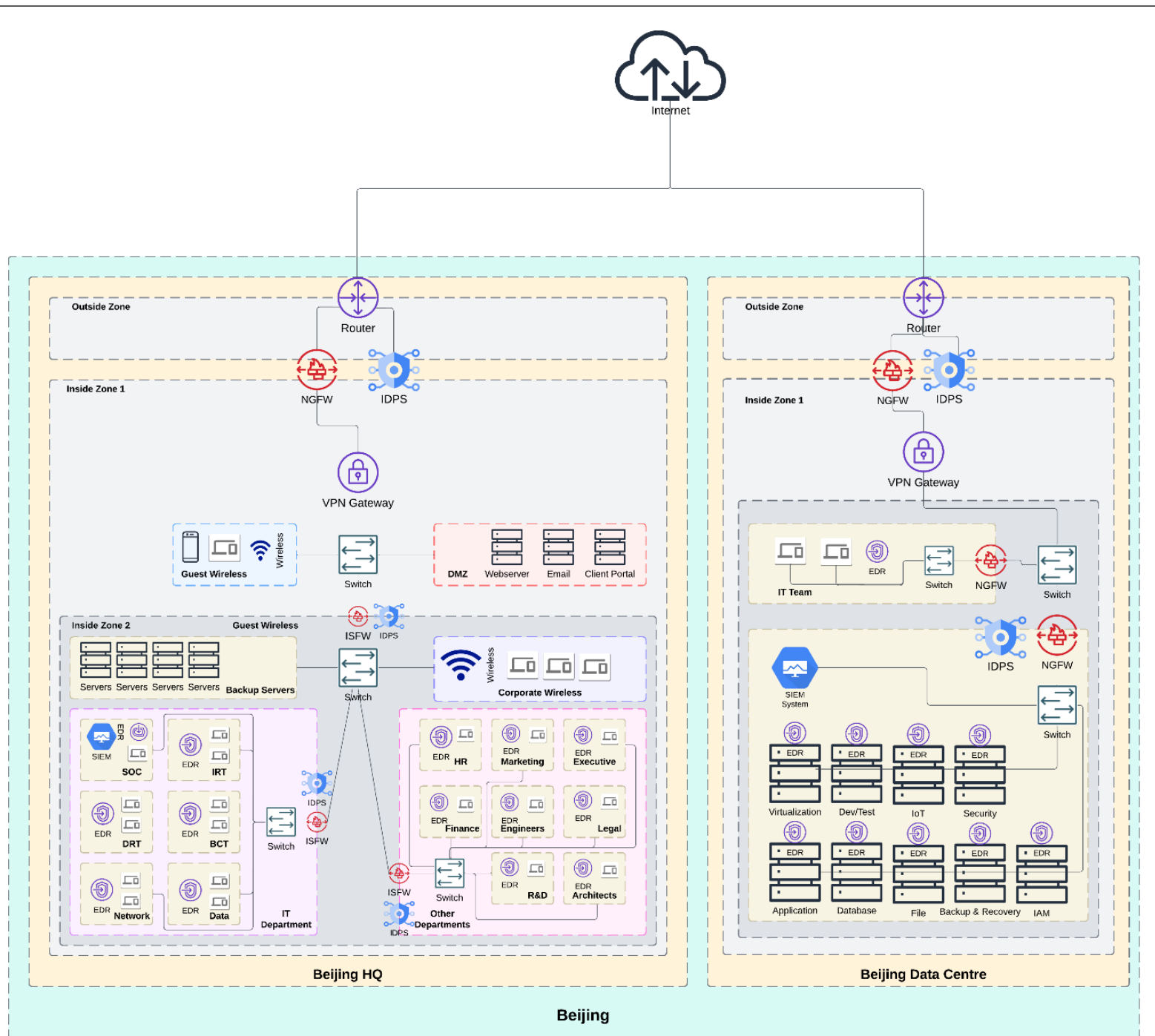
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viii. Network Security Architecture

Following figure covers the security architecture of Liangshan Enterprise (Beijing Headquarters)



1. Headquarters

For the Headquarters, I used the following controls and technologies:

1.1 Gateway Router

The gateway router manages incoming and outgoing internet traffic, ensuring basic routing and forwarding. The gateway router is configured with the access control policies.

1.2 Next-Generation Firewall (NGFW)

NGFW analyzes traffic for malicious patterns, inspects encrypted traffic, and enforces security policies. It is configured inside zone 1 to regulate traffic from external to internal network. Application-layer filtering policies are defined to block known vulnerabilities.

1.3 Internal Segmentation Firewall (ISFW)

ISFW protects network segments from malicious code that makes its way to the internal network. It is configured inside zone 2 to regulate traffic inside the internal network. Application-layer filtering policies are defined to block known vulnerabilities.

1.4 Intrusion Detection and Prevention System (IDPS)

IDPS is placed inline with the NGFW and ISFW to monitor traffic entering and leaving the internal network. It detects and blocks suspicious or malicious activities, including zero-day attacks. It is deployed in "**detect and prevent**" mode for real-time mitigation. Signature-based and anomaly-based detection rules are also enabled.

1.5 VPN Gateway

VPN is placed behind the NGFW, but accessible to authorized remote users from the internet. It ensures encrypted communication for users using remote workstations. It is configured with multi-factor authentication (MFA) for all remote access. Split-tunneling policies are applied to separate internal traffic from public internet use.

1.6 Switches

Switches are positioned at the boundaries of each segment, connecting different departments and segments. They are responsible for segmenting traffic and ensuring efficient communication between internal zones. VLANs are created for departments/zone and visitor networks. Inter-VLAN routing is implemented with security policies using ACLs.

1.7 Access Control Lists (ACLs)

ACLs are applied on core switches and routers managing inter-VLAN and inter-department traffic. They aid in regulating traffic flow based on pre-defined rules. They are configured to allow only specific ports and protocols for communication between departments. They also block traffic between sensitive areas like R&D and visitor Wi-Fi.

1.8 Endpoint Detection and Response (EDR)

EDRs are installed directly on all endpoints (desktops, laptops, and mobile devices). They protect against malware, ransomware, and unauthorized access attempts. Automatic threat detection and response is enabled for real-time security. The policy to regularly update EDR signatures and rules is also enforced by the CISO.

1.9 SIEM (Security Information and Event Management):

SIEM is deployed in the HQ SOC (Security Operations Center). The SIEM solution is responsible for collecting and analyzing logs from all security devices (firewalls, IDPS, EDR) and servers.

2. Regional Office (London, New York & Dubai)

For the regional offices, following controls and technologies were used that were not implemented in the HQ (technologies like gateway router, NGFW, VPN have been implemented in the entire structure of Liangshan:

2.1 Local SIEM Appliance

Local SIEM is deployed in the regional office's server rooms. These SIEM Solutions collect logs from local firewalls, endpoints, and other devices for centralized analysis at HQ. They are configured with **log forwarding** to HQ SIEM for correlation and reporting. Local alerts are also set for immediate response to anomalies.

2.2 Wireless Network Controls

Wireless Network Controls are implemented via secure access points. They are responsible for providing secure Wi-Fi connectivity for employees and visitors. WNCs use WPA3 encryption.

3. Design Studios (Tokyo and Paris)

For the design studios, the following controls and technologies were used that were not implemented in the HQ. The security architecture at the design studios is a bit different from the HQ and regional offices (not much in terms of what technology is used but how the controls are applied)

3.1 Gateway Router

The gateway router is the first entry point for internet traffic. Other functionalities are the same as the HQ gateway router. However, strict routing policies are applied for external

connections. Also, it is configured to allow only secure communication with HQ and regional offices.

3.2 Next-Generation Firewall (NGFW)

NGFW is placed behind the gateway router. It provides advanced filtering for design-related data traffic. Here, deep-packet inspection is enabled for large design file transfers. This firewall also restricts unauthorized protocols and services.

3.3 VPN Gateway

In both the design studios, VPN is integrated with the NGFW. It ensures secure communication with HQ, regional offices, and client systems. Role-based access is applied for design team members.

3.4 Switch

A single centralized switch is used for connecting internal zones (e.g., design teams, visualization labs). It is responsible for managing traffic within the design studio. VLANs are created for design teams, administrative staff, and IoT devices in visualization labs. Inter-VLAN ACLs are enforced as well.

3.5 Data Loss Prevention (DLP)

DLP is applied on all endpoints and internal servers. It prevents unauthorized data transfers and leaks. DLP is responsible for monitoring sensitive data (e.g., project files) being sent outside the network. In case of any suspicious activity, it alerts on and blocks that activity.

3.6 Endpoint Protection:

Endpoint protection is applied on all designer workstations and laptops. It safeguards high-value devices from threats. It is configured to regularly update antivirus definitions and enforce application whitelisting.

3.7 Collaboration Tools Security:

CTS is deployed on BIM platforms and 3D visualization systems. It enables secure file sharing and project collaboration. It encrypts all file transfers between design teams and clients.

4. Data Centers (Beijing and London)

For the data centers, the following controls and technologies were used that were not implemented above. The security architecture for the data centers is similar to that of HQ. However additional security measures have been placed for servers' security.

4.1 Gateway Router:

The gateway router acts as the first entry point for external connections. It is responsible for routing traffic between external networks and the data center. It enforces strict routing policies to restrict traffic from unauthorized sources. Router allows only specific traffic such as VPN connections, replication traffic, and secure file transfers.

4.2 Next-Generation Firewall (NGFW):

NGFW is positioned directly behind the gateway router. It protects the data center perimeter by analyzing traffic for threats. Application-aware filtering is enabled

4.3 Intrusion Detection and Prevention System (IDPS):

Same as the HQ, IDPS is deployed in line with the firewall. Custom rules are implemented for detecting specific threats, such as SQL injection or brute-force attacks.

4.4 VPN Gateway:

The VPN in the data center is integrated with the firewall. It provides secure connections for remote management and inter-data center replication. The VPN has been configured to use IPsec for encryption and authentication. MFA has been enforced for all VPN users.

4.5 Storage Area Network (SAN) Security:

SAN is deployed for data storage and backups. It manages high-speed data transfers within the data center. It is configured to encrypt data in transit using Fiber Channel encryption. SAN uses multi-path input/output (MPIO) for redundancy.

4.6 Endpoint Detection and Response (EDR):

Similar to HQ, EDRs have been installed on servers and administrator devices. They are responsible for protecting against malware and unauthorized access. Threat detection and automated remediation has enabled quick for quick reaction. To reduce noise, the rules are monitored regularly.

4.7 SIEM System:

A centralized SIEM System set up is maintained in the server room. It aggregates logs from all devices and provides real-time analysis. Automated alerts have been set up for anomalies in server and network activity. It is integrated with HQ SIEM for centralized monitoring.